Continuous Integration (CI)

Lecture 3

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Lesson's Agenda

Section 1: CI Concepts

Section 2: GitHub Actions Anatomy

Section 3: Building the pipeline

Section 4: Wrap Up

Section 1: CI Concepts

Life Before CI

The Manual Pain

Developer finishes code ⇨ Manually runs tests
⇨ Manually builds artifacts ⇨ Manually uploads.

Risks:

"I forgot to run tests."

"It works on my machine but fails on the server."

Wasted time waiting for builds.

What is CI ?

The practice of merging code changes frequently (daily) into a shared repository.

Every merge triggers an **Automated Sequence**:

Build the code

Run tests

Report status

The CI “Gatekeeper”

Protecting the Codebase

CI acts as a security guard for the main branch.

Red Build ❌: Tests failed. Merge is blocked.

Green Build ✅: Safe to merge.

Goal: "Fail Fast". Find bugs immediately after committing.

Why GitHub Actions?

Tool Choice

Integrated: Built right into GitHub. No external servers (like Jenkins).

Free: Generous free tier for students and public repos.

Community: Thousands of pre-built "**Actions**" in the Marketplace.

Configuration: Defined as Code (YAML).

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Section 2: GitHub Actions Anatomy

Core Components

How it works ?

Workflow: The automated process (**.yml** file).

Event: What triggers it? (Push, PR).

Job: A set of steps running on a machine.

Step: Individual task within a job.

The Workflow File

YAML file

Stored in **.github/workflows/** directory.

Example: **.github/workflows/ci.yml**.

Must be in the root of your repository.

Triggers (Events)

When to run ?

push: Runs when code is pushed.

pull_request: Runs when a PR is opened/updated
(Critical for Code Review).

workflow_dispatch: Adds a button to run manually.

```yaml
on:
  push:
    branches:
      - main
  pull_request:
    branches:
      - main
```

Runners

The Execution Environment

A **Runner** is a virtual machine hosted by GitHub.

It is a fresh, clean computer created just for your job.

It is destroyed immediately after the job finishes.

Options: ubuntu-latest, windows-latest, macos-latest.

```
runs-on: ubuntu-latest
```

Jobs

Grouping Tasks

A workflow can have multiple jobs (e.g., *test, build, deploy*).

Parallelism: Jobs run in parallel by default.

Dependency: Use needs: test to make deploy wait for test.

Steps

The Task List

Inside a job, steps run sequentially.

Type 1 - Command: run: npm install
(Executes shell command).

Type 2 - Action: uses: actions/checkout@v3
(Uses a pre-made script).

GitHub Actions Marketplace

Don’t reinvent the Wheel

The Marketplace has actions for everything.

Need Node? **actions/setup-node**.

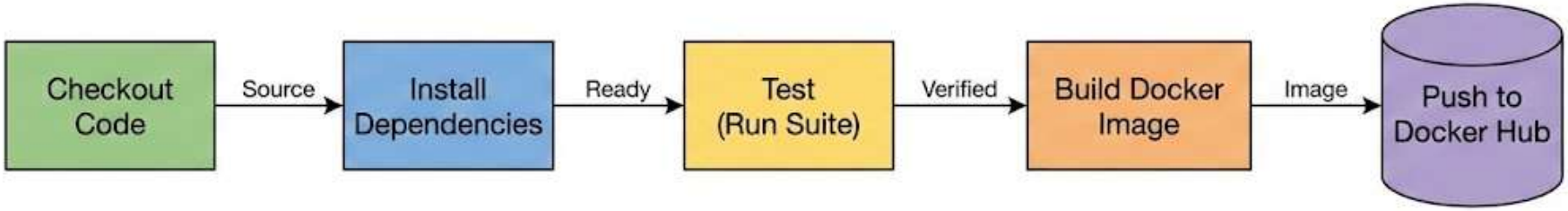
Need Docker? **docker/login-action**.

Need AWS? **aws-actions/configure-aws-credentials**.

Section 3: Building the pipeline

Pipeline Overview

- This is the standard pipeline for modern web apps.
  1. **Checkout:** Downloads the code to the runner.
  2. **Setup Environment:** Prepares the runtime environment (e.g., Node.js).
  3. **Install Dependencies:** Installs the necessary packages using `npm ci`.
  4. **Testing:** Runs the test suite to ensure the code is correct.
  5. **Docker Login:** Logs into Docker to push images.
  6. **Build & Push:** Builds the Docker image and uploads it to the registry.



Step 1 – Checkout

- The Runner is empty.
- This step downloads your Git repository code onto the Runner.
- Without this, you have no code to test!

```yaml
steps:
  - uses: actions/checkout@v3
```

Step 2 – Setup Environment

Setting up Node.js

- Prepares the language runtime.

```yaml
- uses: actions/setup-node@v3
  with:
    node-version: 18
```

Step 3 – Install Dependencies

- **npm ci** (*Clean Install*) is faster and strictly follows **package-lock.json**
- Use this for CI servers instead of npm install.

```
run: npm ci
```

Step 4 – Testing

- Executes your test suite (Jest, Mocha, etc.).
- **The Rule:** If this command returns an error (exit code 1), the **entire workflow stops**.
- This prevents bad code from reaching the build stage.

```
run: npm test
```

Putting It Together: ci.yml

Example CI workflow: build-and-test

- CI runs on every Pull Request to **main**
- **services.db** runs PostgreSQL in the CI runner (no external DB needed)
- Tests connect to **localhost:5432** using environment variables

```yaml
name: CI - Build & Test

on:
  pull_request:
    branches: [ main ]

jobs:
  build-and-test:
    runs-on: ubuntu-latest

    services:
      db:
        image: postgres:15-alpine
        env:
          POSTGRES_USER: postgres
          POSTGRES_PASSWORD: postgres_password
          POSTGRES_DB: tododb
        ports:
          - 5432:5432
        options: >-
          --health-cmd="pg_isready -U postgres"
          --health-interval=10s
          --health-timeout=5s
          --health-retries=5
```

```yaml
steps:
  - name: Checkout code
    uses: actions/checkout@v4

  - name: Setup Node.js
    uses: actions/setup-node@v4
    with:
      node-version: 18

  - name: Install dependencies
    run: npm ci

  - name: Run tests
    env:
      DB_HOST: localhost
      DB_PORT: 5432
      DB_USER: postgres
      DB_PASSWORD: postgres_password
    run: npm test
```

Step 5 – Docker Login

To push an image, we must log in to Docker Hub.

Security Risk: Do NOT put passwords in the YAML file.

Solution: Use GitHub Secrets.

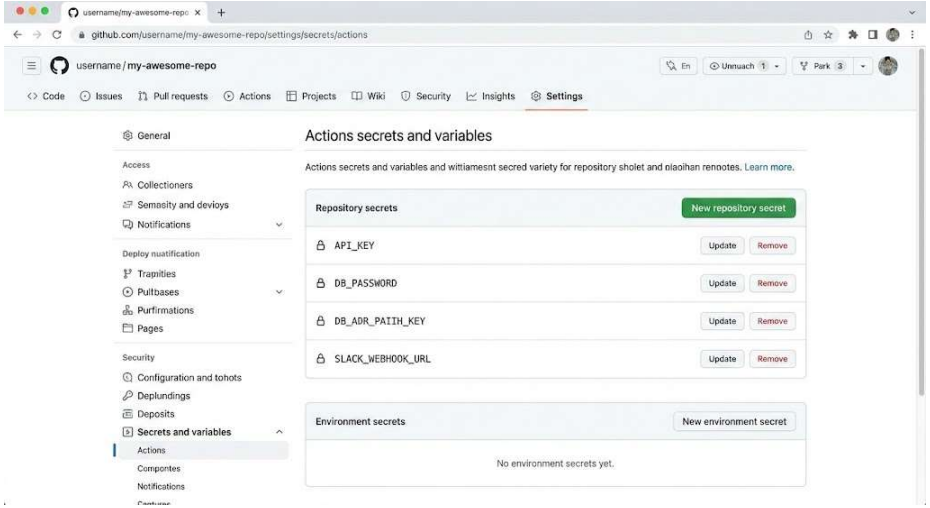
Managing Secrets

GitHub Secrets

Create DOCKER_USERNAME and DOCKER_PASSWORD

Access in YAML: ${{ secrets.DOCKER_PASSWORD }}

GitHub hides these values in logs (***).



Step 6 – Build & Push

- Builds the **Dockerfile**.
- Uploads the image to the registry.

```yaml
- uses: docker/build-push-action@v4
  with:
    push: true
    tags: myuser/myapp:latest
```

Step 5-6: Build & Push Docker Image

Example: build and push Docker image

- Docker Hub credentials must be stored in GitHub Secrets (never in YAML).
- Image name should match the group's repo/image naming convention

```yaml
- name: Log in to Docker Hub
  uses: docker/login-action@v3
  with:
    username: ${{ secrets.DOCKER_HUB_USERNAME }}
    password: ${{ secrets.DOCKER_HUB_PASSWORD }}

- name: Build and push image
  uses: docker/build-push-action@v6
  with:
    push: true
    tags: ${{ secrets.DOCKER_HUB_USERNAME }}/devops-todo:latest
```

Caching (Optimization)

Speeding it up

Downloading **node_modules** every time is slow.

Cache Action: actions/cache can save dependencies between runs.

Faster feedback = Happier developers.

Caching Node Modules in CI

NodeJS dependency caching

- Cache key is based on **package-lock.json**, so cache is invalidated when deps change
- This speed up **npm ci** in repeated runs

```yaml
- name: Cache node_modules
  uses: actions/cache@v4
  with:
    path: |
      ~/.npm
      node_modules
    key: ${{ runner.os }}-node-${{ hashFiles('package-lock.json') }}
    restore-keys: |
      ${{ runner.os }}-node-
```

Artifacts

Sometimes you want to save the output
(e.g., a .jar file or a compiled website).

Use **actions/upload-artifact** to store files for download later.

Section 4: Wrap Up

Branch Protection Rules

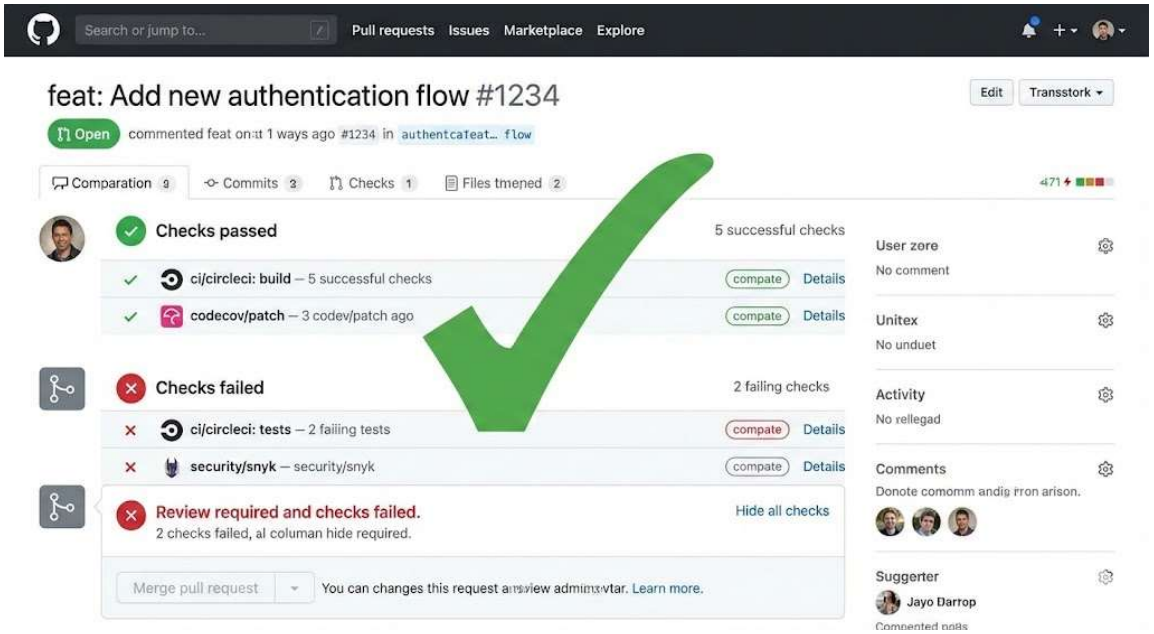
Enforcing CI

1. Go to GitHub Repo Settings ⇨ Branches.
2. Enable "*Require status checks to pass before merging*".
3. Select your *build-and-test* job.
4. Now, nobody can merge broken code!

Status Indicators

Visibility

- This provides immediate feedback to the developer.



Debugging Failures

- Click on the "**Details**" button in the PR.
- Read the console logs.
- Common errors:
  - Syntax error in code.
  - Test logic failure.
  - Missing Secret.
  - YAML indentation error.

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Local Testing

Test Locally First

Don't treat the CI server as your testing ground.

Run npm test and docker build on your laptop *before* pushing.

"Respect the CI resources."

Adding Deploy Job (CD Preview)

Deploy job in CI

- **needs: build-and-test** ensures CD only runs if CI passed.
- **if: github.ref == 'refs/heads/main'** prevents deployments from PR branches.

```yaml
deploy:
  needs: build-and-test
  if: github.ref == 'refs/heads/main'
  runs-on: ubuntu-latest

  steps:
    - name: Deploy via SSH
      uses: appleboy/ssh-action@v1
      with:
        host: ${{ secrets.HOST }}
        username: ${{ secrets.USERNAME }}
        key: ${{ secrets.SSH_PRIVATE_KEY }}
        script: |
          cd /path/to/your-project
          git pull origin main
          docker-compose pull
          docker-compose up -d
```

Broken Test Example

Fail ⇨ Fix ⇨ Pass

- First, intentionally make the test *wrong* so CI fails
- Then fix the application and/or test so the pipeline turns *green*

```
// tests/todos.test.js
it('POST /api/todos should fail without title', async () => {
  const res = await request(app)
    .post('/api/todos')
    .send({})
    .expect(400); // EXPECT 400 (Bad Request), not 500

  expect(res.body.error).toBe('Title is required');
});
```

Key Takeaways

- CI automates integration.
- GitHub Actions uses YAML workflows.
- Secrets keep passwords safe.
- Status Checks protect the main branch.

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